

INTRODUCTION

Syntax is the study of the principles and processes by which sentences are constructed in particular languages. Syntactic investigation of a given language has as its goal the construction of a grammar that can be viewed as a device of some sort for producing the sentences of the language under analysis. More generally, linguists must be concerned with the problem of determining the fundamental underlying properties of successful grammars. The ultimate outcome of these investigations should be a theory of linguistic structure in which the descriptive devices utilized in particular grammars are presented and studied abstractly, with no specific reference to particular languages. One function of this theory is to provide a general method for selecting a grammar for each language, given a corpus of sentences of this language.

The central notion in linguistic theory is that of "linguistic level." A linguistic level, such as phonemics, morphology, phrase structure, is essentially a set of descriptive devices that are made available for the construction of grammars; it constitutes a certain method for representing utterances. We can determine the adequacy of a linguistic theory by developing rigorously and precisely the form of grammar corresponding to the set of levels contained within this theory, and then investigating the possibility of constructing simple and revealing grammars of this form for natural languages. We shall study several different conceptions of linguistic structure in this manner, considering a succession of linguistic levels of increasing complexity which correspond to more and more powerful modes of grammatical description; and we shall attempt to show that linguistic theory must contain at least these levels if it is to

provide, in particular, a satisfactory grammar of English. Finally, we shall suggest that this purely formal investigation of the structure of language has certain interesting implications for semantic studies.¹

¹ The motivation for the particular orientation of the research reported here is discussed below in § 6.

THE INDEPENDENCE OF GRAMMAR

2.1 From now on I will consider a *language* to be a set (finite or infinite) of sentences, each finite in length and constructed out of a finite set of elements. All natural languages in their spoken or written form are languages in this sense, since each natural language has a finite number of phonemes (or letters in its alphabet) and each sentence is representable as a finite sequence of these phonemes (or letters), though there are infinitely many sentences. Similarly, the set of 'sentences' of some formalized system of mathematics can be considered a language. The fundamental aim in the linguistic analysis of a language *L* is to separate the *grammatical* sequences which are the sentences of *L* from the *ungrammatical* sequences which are not sentences of *L* and to study the structure of the grammatical sequences. The grammar of *L* will thus be a device that generates all of the grammatical sequences of *L* and none of the ungrammatical ones. One way to test the adequacy of a grammar proposed for *L* is to determine whether or not the sequences that it generates are actually grammatical, i.e., acceptable to a native speaker, etc. We can take certain steps towards providing a behavioral criterion for grammaticalness so that this test of adequacy can be carried out. For the purposes of this discussion, however, suppose that we assume intuitive knowledge of the grammatical sentences of English and ask what sort of grammar will be able to do the job of producing these in some effective and illuminating way. We thus face a familiar task of explication of some intuitive concept — in this case, the concept "grammatical in English," and more generally, the concept "grammatical."

Notice that in order to set the aims of grammar significantly it is sufficient to assume a partial knowledge of sentences and non-

sentences. That is, we may assume for this discussion that certain sequences of phonemes are definitely sentences, and that certain other sequences are definitely non-sentences. In many intermediate cases we shall be prepared to let the grammar itself decide, when the grammar is set up in the simplest way so that it includes the clear sentences and excludes the clear non-sentences. This is a familiar feature of explication.¹ A certain number of clear cases, then, will provide us with a criterion of adequacy for any particular grammar. For a single language, taken in isolation, this provides only a weak test of adequacy, since many different grammars may handle the clear cases properly. This can be generalized to a very strong condition, however, if we insist that the clear cases be handled properly for *each* language by grammars all of which are constructed by the same method. That is, each grammar is related to the corpus of sentences in the language it describes in a way fixed in advance for all grammars by a given linguistic theory. We then have a very strong test of adequacy for a linguistic theory that attempts to give a general explanation for the notion "grammatical sentence" in terms of "observed sentence," and for the set of grammars constructed in accordance with such a theory. It is furthermore a reasonable requirement, since we are interested not only in particular languages, but also in the general nature of Language. There is a great deal more that can be said about this crucial topic, but this would take us too far afield. Cf. § 6.

2.2 On what basis do we actually go about separating grammatical sequences from ungrammatical sequences? I shall not attempt to

¹ Cf., for example, N. Goodman, *The structure of appearance* (Cambridge, 1951), pp. 5-6. Notice that to meet the aims of grammar, given a linguistic theory, it is sufficient to have a partial knowledge of the sentences (i.e., a corpus) of the language, since a linguistic theory will state the relation between the set of observed sentences and the set of grammatical sentences; i.e., it will define "grammatical sentence" in terms of "observed sentence," certain properties of the observed sentences, and certain properties of grammars. To use Quine's formulation, a linguistic theory will give a general explanation for what 'could' be in language on the basis of "what *is* plus *simplicity* of the laws whereby we describe and extrapolate what is". (W. V. Quine, *From a logical point of view* [Cambridge, 1953], p. 54). Cf. § 6.1.

give a complete answer to this question here (cf. §§ 6.7), but I would like to point out that several answers that immediately suggest themselves could not be correct. First, it is obvious that the set of grammatical sentences cannot be identified with any particular corpus of utterances obtained by the linguist in his field work. Any grammar of a language will *project* the finite and somewhat accidental corpus of observed utterances to a set (presumably infinite) of grammatical utterances. In this respect, a grammar mirrors the behavior of the speaker who, on the basis of a finite and accidental experience with language, can produce or understand an indefinite number of new sentences. Indeed, any explication of the notion "grammatical in L" (i.e., any characterization of "grammatical in L" in terms of "observed utterance of L") can be thought of as offering an explanation for this fundamental aspect of linguistic behavior.

2.3 Second, the notion "grammatical" cannot be identified with "meaningful" or "significant" in any semantic sense. Sentences (1) and (2) are equally nonsensical, but any speaker of English will recognize that only the former is grammatical.

- (1) Colorless green ideas sleep furiously.
- (2) Furiously sleep ideas green colorless.

Similarly, there is no semantic reason to prefer (3) to (5) or (4) to (6), but only (3) and (4) are grammatical sentences of English.

- (3) have you a book on modern music?
- (4) the book seems interesting.
- (5) read you a book on modern music?
- (6) the child seems sleeping.

Such examples suggest that any search for a semantically based definition of "grammaticalness" will be futile. We shall see, in fact, in § 7, that there are deep structural reasons for distinguishing (3) and (4) from (5) and (6); but before we are able to find an explanation for such facts as these we shall have to carry the theory of syntactic structure a good deal beyond its familiar limits.

2.4 Third, the notion "grammatical in English" cannot be identi-

fied in any way with the notion "high order of statistical approximation to English." It is fair to assume that neither sentence (1) nor (2) (nor indeed any part of these sentences) has ever occurred in an English discourse. Hence, in any statistical model for grammaticalness, these sentences will be ruled out on identical grounds as equally 'remote' from English. Yet (1), though nonsensical, is grammatical, while (2) is not. Presented with these sentences, a speaker of English will read (1) with a normal sentence intonation, but he will read (2) with a falling intonation on each word; in fact, with just the intonation pattern given to any sequence of unrelated words. He treats each word in (2) as a separate phrase. Similarly, he will be able to recall (1) much more easily than (2), to learn it much more quickly, etc. Yet he may never have heard or seen any pair of words from these sentences joined in actual discourse. To choose another example, in the context "I saw a fragile—," the words "whale" and "of" may have equal (i.e., zero) frequency in the past linguistic experience of a speaker who will immediately recognize that one of these substitutions, but not the other, gives a grammatical sentence. We cannot, of course, appeal to the fact that sentences such as (1) 'might' be uttered in some sufficiently far-fetched context, while (2) would never be, since the basis for this differentiation between (1) and (2) is precisely what we are interested in determining.

Evidently, one's ability to produce and recognize grammatical utterances is not based on notions of statistical approximation and the like. The custom of calling grammatical sentences those that "can occur", or those that are "possible", has been responsible for some confusion here. It is natural to understand "possible" as meaning "highly probable" and to assume that the linguist's sharp distinction between grammatical and ungrammatical² is motivated by a feeling that since the 'reality' of language is too complex to be described completely, he must content himself with a schematized

² Below we shall suggest that this sharp distinction may be modified in favor of a notion of levels of grammaticalness. But this has no bearing on the point at issue here. Thus (1) and (2) will be at different levels of grammaticalness even if (1) is assigned a lower degree of grammaticalness than, say, (3) and (4); but they will be at the same level of statistical remoteness from English. The same is true of an indefinite number of similar pairs.

version replacing "zero probability, and all extremely low probabilities, by *impossible*, and all higher probabilities by *possible*."³ We see, however, that this idea is quite incorrect, and that a structural analysis cannot be understood as a schematic summary developed by sharpening the blurred edges in the full statistical picture. If we rank the sequences of a given length in order of statistical approximation to English, we will find both grammatical and ungrammatical sequences scattered throughout the list; there appears to be no particular relation between order of approximation and grammaticality. Despite the undeniable interest and importance of semantic and statistical studies of language, they appear to have no direct relevance to the problem of determining or characterizing the set of grammatical utterances. I think that we are forced to conclude that grammar is autonomous and independent of meaning, and that probabilistic models give no particular insight into some of the basic problems of syntactic structure.⁴

³ C. F. Hockett, *A manual of phonology* (Baltimore, 1955), p. 10.

⁴ We return to the question of the relation between semantics and syntax in §§ 8, 9, where we argue that this relation can only be studied after the syntactic structure has been determined on independent grounds. I think that much the same thing is true of the relation between syntactic and statistical studies of language. Given the grammar of a language, one can study the use of the language statistically in various ways; and the development of probabilistic models for the use of language (as distinct from the syntactic structure of language) can be quite rewarding. Cf. B. Mandelbrot, "Structure formelle des textes et communication: deux études," *Word* 10.1-27 (1954); H. A. Simon, "On a class of skew distribution functions," *Biometrika* 42.425-40 (1955).

One might seek to develop a more elaborate relation between statistical and syntactic structure than the simple order of approximation model we have rejected. I would certainly not care to argue that any such relation is unthinkable, but I know of no suggestion to this effect that does not have obvious flaws. Notice, in particular, that for any n , we can find a string whose first n words may occur as the beginning of a grammatical sentence S_1 and whose last n words may occur as the ending of some grammatical sentence S_2 , but where S_1 must be distinct from S_2 . For example, consider the sequences of the form "the man who ... are here," where ... may be a verb phrase of arbitrary length. Notice also that we can have new but perfectly grammatical sequences of word classes, e.g., a sequence of adjectives longer than any ever before produced in the context "I saw a — house." Various attempts to explain the grammatical-ungrammatical distinction, as in the case of (1), (2), on the basis of frequency of sentence type, order of approximation of word class sequences, etc., will run afoul of numerous facts like these.

ON THE GOALS OF LINGUISTIC THEORY

6.1 In §§ 3, 4 two models of linguistic structure were developed: a simple communication theoretic model and a formalized version of immediate constituent analysis. Each was found to be inadequate, and in § 5 I suggested a more powerful model combining phrase structure and grammatical transformations that might remedy these inadequacies. Before going on to explore this possibility, I would like to clarify certain points of view that underlie the whole approach of his study.

Our fundamental concern throughout this discussion of linguistic structure is the problem of justification of grammars. A grammar of the language *L* is essentially a theory of *L*. Any scientific theory is based on a finite number of observations, and it seeks to relate the observed phenomena and to predict new phenomena by constructing general laws in terms of hypothetical constructs such as (in physics, for example) "mass" and "electron." Similarly, a grammar of English is based on a finite corpus of utterances (observations), and it will contain certain grammatical rules (laws) stated in terms of the particular phonemes, phrases, etc., of English (hypothetical constructs). These rules express structural relations among the sentences of the corpus and the indefinite number of sentences generated by the grammar beyond the corpus (predictions). Our problem is to develop and clarify the criteria for selecting the correct grammar for each language, that is, the correct theory of this language.

Two types of criteria were mentioned in § 2.1. Clearly, every grammar will have to meet certain *external conditions of adequacy*; e.g., the sentences generated will have to be acceptable to the native

speaker. In § 8 we shall consider several other external conditions of this sort. In addition, we pose a *condition of generality* on grammars; we require that the grammar of a given language be constructed in accordance with a specific theory of linguistic structure in which such terms as "phoneme" and "phrase" are defined independently of any particular language.¹ If we drop either the external conditions or the generality requirement, there will be no way to choose among a vast number of totally different 'grammars,' each compatible with a given corpus. But, as we observed in § 2.1, these requirements jointly give us a very strong test of adequacy for a general theory of linguistic structure and the set of grammars that it provides for particular languages.

Notice that neither the general theory nor the particular grammars are fixed for all time, in this view. Progress and revision may come from the discovery of new facts about particular languages, or from purely theoretical insights about organization of linguistic data — that is, new models for linguistic structure. But there is also no circularity in this conception. At any given time we can attempt to formulate as precisely as possible both the general theory and the set of associated grammars that must meet the empirical, external conditions of adequacy.

We have not yet considered the following very crucial question: What is the relation between the general theory and the particular grammars that follow from it? In other words, what sense can we give to the notion "follow from," in this context? It is at this point that our approach will diverge sharply from many theories of linguistic structure.

The strongest requirement that could be placed on the relation between a theory of linguistic structure and particular grammars is that the theory must provide a practical and mechanical method for

¹ I presume that these two conditions are similar to what Hjelmslev has in mind when he speaks of the *appropriateness* and *arbitrariness* of linguistic theory. Cf. L. Hjelmslev, *Prolegomena to a theory of language* = *Memoir 7, Indiana University Publications Anthropology and Linguistics* (Baltimore, 1953), p. 8. See also Hockett's discussion of "metacriteria" for linguistics ("Two models of grammatical description," *Linguistics Today*, *Word* 10.232-3) in this connection.

actually constructing the grammar, given a corpus of utterances. Let us say that such a theory provides us with a *discovery procedure* for grammars.

A weaker requirement would be that the theory must provide a practical and mechanical method for determining whether or not a grammar proposed for a given corpus is, in fact, the best grammar of the language from which this corpus is drawn. Such a theory, which is not concerned with the question of *how* this grammar was constructed, might be said to provide a *decision procedure* for grammars.

An even weaker requirement would be that given a corpus and given two proposed grammars G_1 and G_2 , the theory must tell us which is the better grammar of the language from which the corpus is drawn. In this case we might say that the theory provides an *evaluation procedure* for grammars.

These theories can be represented graphically in the following manner.

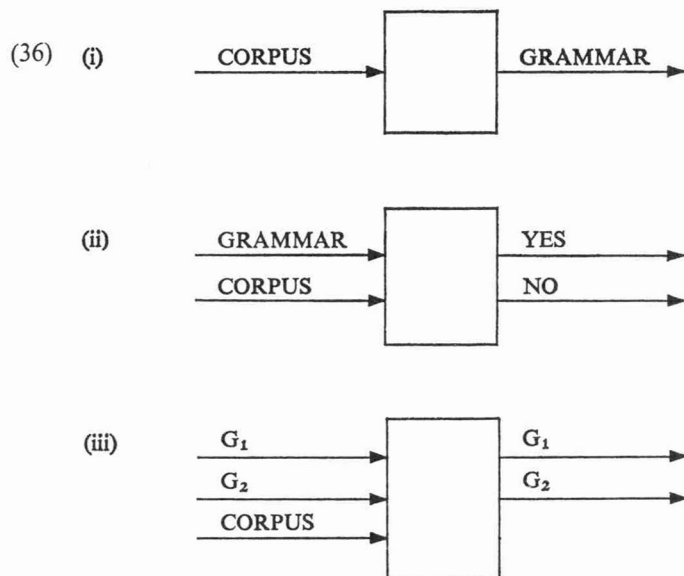


Figure (36i) represents a theory conceived as a machine with a corpus as its input and a grammar as its output; hence, a theory that provides a discovery procedure. (36ii) is a device with a grammar and a corpus as its inputs, and the answers "yes" or "no" as its outputs, as the grammar is or is not the correct one; hence, it represents a theory that provides a decision procedure for grammars. (36iii) represents a theory with grammars G_1 and G_2 and a corpus as inputs, and the more preferable of G_1 and G_2 as output; hence a theory that provides an evaluation procedure for grammars.²

The point of view adopted here is that it is unreasonable to demand of linguistic theory that it provide anything more than a practical evaluation procedure for grammars. That is, we adopt the weakest of the three positions described above. As I interpret most of the more careful proposals for the development of linguistic theory,³ they attempt to meet the strongest of these three requirements. That is, they attempt to state methods of analysis that an investigator might actually use, if he had the time, to construct a grammar of a language directly from the raw data. I think that it is very questionable that this goal is attainable in any interesting way,

² The basic question at issue is not changed if we are willing to accept a small set of correct grammars instead of a single one.

³ For example, B. Bloch, "A set of postulates for phonemic analysis," *Language* 24.3-46 (1948); N. Chomsky, "Systems of syntactic analysis," *Journal of Symbolic Logic* 18.242-56 (1953); Z. S. Harris, "From phoneme to morpheme," *Language* 31.190-222 (1955); idem, *Methods in structural linguistics* (Chicago, 1951); C. F. Hockett, "A formal statement of morphemic analysis," *Studies in Linguistics* 10.27-39 (1952); idem, "Problems of morphemic analysis," *Language* 23.321-43 (1947); R. S. Wells, "Immediate constituents," *Language* 23.81-117 (1947); and many other works. Although discovery procedures are the explicit goal of these works, we often find on careful examination that the theory that has actually been constructed furnishes no more than an evaluation procedure for grammars. For example, Hockett states his aim in "A formal statement of morphemic analysis" as the development of "formal procedures by which one can work from scratch to a complete description of the pattern of a language" (p. 27); but what he actually does is describe some of the formal properties of a morphological analysis and then propose a "criterion whereby the relative efficiency of two possible morphic solutions can be determined; with that, we can choose the maximally efficient possibility, or, arbitrarily, any one of those which are equally efficient but more efficient than all others" (p. 29).

and I suspect that any attempt to meet it will lead into a maze of more and more elaborate and complex analytic procedures that will fail to provide answers for many important questions about the nature of linguistic structure. I believe that by lowering our sights to the more modest goal of developing an evaluation procedure for grammars we can focus attention more clearly on really crucial problems of linguistic structure and we can arrive at more satisfying answers to them. The correctness of this judgment can only be determined by the actual development and comparison of theories of these various sorts. Notice, however, that the weakest of these three requirements is still strong enough to guarantee significance for a theory that meets it. There are few areas of science in which one would seriously consider the possibility of developing a general, practical, mechanical method for choosing among several theories, each compatible with the available data.

In the case of each of these conceptions of linguistic theory we have qualified the characterization of the type of procedure by the word "practical". This vague qualification is crucial for an empirical science. Suppose, for example, that we were to evaluate grammars by measuring some such simple property as length. Then it would be correct to say that we have a practical evaluation procedure for grammars, since we could count the number of symbols they contain; and it would also be literally correct to say that we have a discovery procedure, since we can order all sequences of the finite number of symbols from which grammars are constructed in terms of length, and we can test each of these sequences to see if it is a grammar, being sure that after some finite amount of time we shall find the shortest sequence that qualifies. But this is not the type of discovery procedure that is contemplated by those who are attempting to meet the strong requirement discussed above.

Suppose that we use the word "simplicity" to refer to the set of formal properties of grammars that we shall consider in choosing among them. Then there are three main tasks in the kind of program for linguistic theory that we have suggested. First, it is necessary to state precisely (if possible, with operational, behavioral tests) the external criteria of adequacy for grammars. Second, we

must characterize the form of grammars in a general and explicit way so that we can actually propose grammars of this form for particular languages. Third, we must analyze and define the notion of simplicity that we intend to use in choosing among grammars all of which are of the proper form. Completion of the latter two tasks will enable us to formulate a general theory of linguistic structure in which such notions as "phoneme in L", "phrase in L", "transformation in L" are defined for an arbitrary language L in terms of physical and distributional properties of utterances of L and formal properties of grammars of L.⁴ For example, we shall define the set of phonemes of L as a set of elements which have certain physical and distributional properties, and which appear in the simplest grammar for L. Given such a theory, we can attempt to construct grammars for actual languages, and we can determine whether or not the simplest grammars that we can find (i.e., the grammars that the general theory compels us to choose) meet the external conditions of adequacy. We shall continue to revise our notions of simplicity and our characterization of the form of grammars until the grammars selected by the theory do meet the external conditions.⁵ Notice that this theory may not tell us, in any practical way, how to actually go about constructing the grammar of a given language from a corpus. But it must tell us how to evaluate such a grammar; it must thus enable us to choose between two proposed grammars.

In the preceding sections of this study we have been concerned with the second of these three tasks. We have assumed that the set of grammatical sentences of English is given and that we have some notion of simplicity, and we have tried to determine what sort of grammar will generate exactly the grammatical sentences in some simple way. To formulate this goal in somewhat different terms,

⁴ Linguistic theory will thus be formulated in a metalanguage to the language in which grammars are written — a metametalanguage to any language for which a grammar is constructed.

⁵ We may in fact revise the criteria of adequacy, too, in the course of research. That is, we may decide that certain of these tests do not apply to grammatical phenomena. The subject matter of a theory is not completely determined in advance of investigation. It is partially determined by the possibility of giving an organized and systematic account of some range of phenomena.

we remarked above that one of the notions that must be defined in general linguistic theory is "sentence in L." Entering into the definitions will be such terms as "observed utterance in L", "simplicity of the grammar of L," etc. This general theory is accordingly concerned with clarifying the relation between the set of grammatical sentence and the set of observed sentences. Our investigation of the structure of the former set is a preparatory study, proceeding from the assumption that before we can characterize this relation clearly, we will have to know a great deal more about the formal properties of each of these sets.

In § 7 below, we shall continue to investigate the relative complexity of various ways of describing English structure. In particular, we shall be concerned with the question of whether the whole grammar is simplified if we consider a certain class of sentences to be kernel sentences or if we consider them to be derived by transformation. We thus arrive at certain decisions about the structure of English. In § 8 we shall argue that there is independent evidence in favor of our method for selecting grammars. That is, we shall try to show that the simpler grammars meet certain external conditions of adequacy while the more complex grammars that embody different decisions about assignment of sentences to the kernel, etc., fail these conditions. These results can be no more than suggestive, however, until we give a rigorous account of the notion of simplicity employed. I think that such an account can be given, but this would go beyond the scope of the present monograph. Nevertheless, it should be fairly clear that under any reasonable definition of "simplicity of grammar", most of the decisions about relative complexity that we reach below will stand.⁶

Notice that simplicity is a *systematic* measure; the only ultimate

⁶ See my *The logical structure of linguistic theory* for discussion of methods for evaluating grammars in terms of formal properties of simplicity.

We are not, incidentally, denying the usefulness of even partially adequate discovery procedures. They may provide valuable hints to the practicing linguist or they may lead to a small set of grammars that can then be evaluated. Our main point is that a linguistic theory should not be identified with a manual of useful procedures, nor should it be expected to provide mechanical procedures for the discovery of grammars.

criterion in evaluation is the simplicity of the whole system. In discussing particular cases, we can only indicate how one or another decision will affect the over-all complexity. Such validation can only be tentative, since by simplifying one part of the grammar we may complicate other parts. It is when we find that simplification of one part of the grammar leads to corresponding simplification of other parts that we feel that we are really on the right track. Below, we shall try to show that the simplest transformational analysis of one class of sentences does quite frequently clear the way to a simpler analysis of other classes.

In short, we shall never consider the question of how one might have arrived at the grammar whose simplicity is being determined; e.g., how one might have discovered the analysis of the verb phrase presented in § 5.3. Question of this sort are not relevant to the program of research that we have outlined above. One may arrive at a grammar by intuition, guess-work, all sorts of partial methodological hints, reliance on past experience, etc. It is no doubt possible to give an organized account of many useful procedures of analysis, but it is questionable whether these can be formulated rigorously, exhaustively and simply enough to qualify as a practical and mechanical discovery procedure. At any rate, this problem is not within the scope of our investigations here. Our ultimate aim is to provide an objective, non-intuitive way to evaluate a grammar once presented, and to compare it with other proposed grammars. We are thus interested in describing the form of grammars (equivalently, the nature of linguistic structure) and investigating the empirical consequences of adopting a certain model for linguistic structure, rather than in showing how, in principle, one might have arrived at the grammar of a language.

6.2 Once we have disclaimed any intention of finding a practical discovery procedure for grammars, certain problems that have been the subject of intense methodological controversy simply do not arise. Consider the problem of interdependence of levels. It has been correctly pointed out that if morphemes are defined in terms of phonemes, and, simultaneously, morphological considerations are

considered relevant to phonemic analysis, then linguistic theory may be nullified by a real circularity. However, interdependence of levels does not necessarily lead to circularity. In this case, for example, we can define "tentative phoneme set" and "tentative morpheme set" independently and we can develop a relation of compatibility that holds between tentative phoneme sets and tentative morpheme sets. We can then define a pair of a phoneme set and a morpheme set for a given language as a compatible pair of a tentative phoneme set and a tentative morpheme set. Our compatibility relation may be framed partially in terms of simplicity considerations; that is, we may define the phonemes and morphemes of a language as the tentative phonemes and morphemes which, among other things, jointly lead to the simplest grammar. This gives us a perfectly straightforward way to define interdependent levels with no circularity. Of course, it does not tell us how to *find* the phonemes and morphemes in a direct, mechanical way. But no other phonemic or morphological theory really meets this strong requirement either, and there is little reason to believe that it can be met significantly. In any event, when we lower our aims to the development of an evaluation procedure, there remains little motivation for any objection to mixing of levels, and there is no difficulty in avoiding circularity in the definition of interdependent levels.⁷

⁷ See Z. S. Harris, *Methods in structural linguistics* (Chicago, 1951) (e.g., Appendix to 7.4, Appendix to 8.2, chapters 9, 12) for examples of procedures which lead to interdependent levels. I think that Fowler's objections to Harris' morphological procedures (cf., *Language* 28.504-9 [1952]) can be met without difficulty by a noncircular formulation of the type just proposed. Cf. C. F. Hockett, *A manual of phonology* = *Memoir 11, Indiana University Publications in Anthropology and Linguistics* (Baltimore, 1955); idem, "Two fundamental problems in phonemics," *Studies in Linguistics* 7.33 (1949); R. Jakobson, "The phonemic and grammatical aspects of language and their interrelation," *Proceedings of the Sixth International Congress of Linguists* 5-18 (Paris, 1948); K. L. Pike, "Grammatical prerequisites to phonemic analysis," *Word* 3.155-72 (1947); idem, "More on grammatical prerequisites," *Word* 8.106-21 (1952), for further discussion on interdependence of levels. Also N. Chomsky, M. Halle, F. Lukoff, "On accent and juncture in English," *For Roman Jakobson* ('s-Gravenhage, 1956), 65-80.

Bar-Hillel has suggested in "Logical syntax and semantics," *Language* 30.230-7 (1954) that Pike's proposals can be formalized without the circularity

Many problems of morphemic analysis also receive quite simple solutions if we adopt the general framework outlined above. In attempting to develop discovery procedures for grammars we are naturally led to consider morphemes as classes of sequences of phonemes, i.e., as having actual phonemic 'content' in an almost literal sense. This leads to trouble in such well-known cases as English "took" /tuk/, where it is difficult without artificiality to associate any part of this word with the past tense morpheme which appears as /t/ in "walked" /wɔkt/, as /d/ in "framed" /freymd/, etc. We can avoid all such problems by regarding morphology and phonology as two distinct but interdependent levels of representation, related in the grammar by morphophonemic rules such as (19). Thus "took" is represented on the morphological level as *take + past* just as "walked" is represented as *walk + past*. The morphophonemic rules (19 ii), (19 v), respectively, carry these strings of morphemes into /tuk/, /wɔkt/. The only difference between the two cases is that (19 v) is a much more general rule than (19 ii).⁸ If we give up the idea that higher levels are literally constructed out of

that many sense in them by the use of recursive definitions. He does not pursue this suggestion in any detail, and my own feeling is that success along these lines is unlikely. Moreover, if we are satisfied with an evaluation procedure for grammars, we can construct interdependent levels with only direct definitions, as we have just seen.

The problem of interdependence of phonemic and morphemic levels must not be confused with the question of whether morphological information is required to read a phonemic transcription. Even if morphological considerations are considered relevant to determining the phonemes of a language, it may still be the case that the phonemic transcription provides complete 'reading' rules with no reference to other levels. Cf. N. Chomsky, M. Halle, F. Lukoff, "On accent and juncture in English," *For Roman Jakobson* ('s-Gravenhage, 1956), 65-80, for discussion and examples.

⁸ Hockett gives a very clear presentation of this approach to levels in *A manual of phonology* (1955), p. 15. In "Two models of grammatical description," *Linguistics Today*, Word 10.210-33 (1954), Hockett rejected a solution very much like the one we have just proposed on the grounds that "*took* and *take* are partly similar in phonemic shape just as are *baked* and *bake*, and similar in meaning also in the same way; this fact should not be obscured" (p. 224). But the similarity in meaning is not obscured in our formulation, since the morpheme *past* appears in the morphemic representation of both "took" and "baked." And the similarity in phonemic shape can be brought out in the actual

lower level elements, as I think we must, then it becomes much more natural to consider even such abstract systems of representation as transformational structure (where each utterance is represented by the sequence of transformations by which it is derived from a terminal string of the phrase structure grammar) as constituting a linguistic level.

We are not actually forced to give up hope of finding a practical discovery procedure by adopting either the view that levels are interdependent, or the conception of linguistic levels as abstract systems of representation related only by general rules. Nevertheless, I think it is unquestionable that opposition to mixing levels, as well as the idea that each level is literally constructed out of lower level elements, has its origin in the attempt to develop a discovery procedure for grammars. If we renounce this goal and if we distinguish clearly between a manual of suggestive and helpful procedures and a theory of linguistic structure, then there is little reason for maintaining either of these rather dubious positions.

There are many other commonly held views that seem to lose much of their appeal if we formulate our goals in the manner suggested above. Thus it is sometimes argued that work on syntactic theory is premature at this time in view of the fact that many of the problems that arise on the lower levels of phonemics and morphology are unsolved. It is quite true that the higher levels of linguistic description depend on results obtained at the lower levels. But there is also a good sense in which the converse is true. For example, we have seen above that it would be absurd, or even hopeless, to state principles of sentence construction in terms of phonemes or morphemes, but only the development of such higher levels as phrase structure indicates that this futile task need not be

formulation of the morphophonemic rule that carries *take + past* into /tuk/. We will no doubt formulate this rule as

ey → u in the context t - k + *past*

in the actual morphophonemic statement. This will allow us to simplify the grammar by a generalization that will bring out the parallel between "take"- "took," "shake"- "shook," "forsake"- "forsook," and more generally, "stand"- "stood," etc.

undertaken on lower levels.⁹ Similarly, we have argued that description of sentence structure by constituent analysis will be unsuccessful, if pushed beyond certain limits. But only the development of the still more abstract level of transformations can prepare the way for the development of a simpler and more adequate technique of constituent analysis with narrower limits. The grammar of a language is a complex system with many and varied interconnections between its parts. In order to develop one part of grammar thoroughly, it is often useful, or even necessary, to have some picture of the character of a completed system. Once again, I think that the notion that syntactic theory must await the solution of problems of phonology and morphology is completely untenable whether or not one is concerned with the problem of discovery procedures, but I think it has been nurtured by a faulty analogy between the order of development of linguistic theory and the presumed order of operations in discovery of grammatical structure.

⁹ See N. Chomsky, M. Halle, F. Lukoff, "On accent and juncture in English," *For Roman Jakobson* ('s-Gravenhage, 1956), 65-80, for a discussion of the possibility that considerations on all higher levels, including morphology, phrase structure, and transformations, are relevant to the selection of a phonemic analysis.